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Quiz 1: Scientific inferences

1: Mark all correct descriptions that apply to inductive inferences.

- [ ] Inductive inferences are typically based on unfounded assumptions.
- [ ] Inductive inferences are such that if they are valid, the conclusion must be true if the premises are true.
- [x] Inductive inferences go beyond information we already have, thus they amplify our knowledge when they are used.
- [ ] Inductive inferences rearrange current knowledge in such a way that they merely explicate what we already know.

2: Mark all correct descriptions that apply to deductive inferences.

- [ ] Deductive inferences are typically based on unfounded assumptions.
- [x] Deductive inferences rearrange current knowledge in such a way that they explicate what we already know, without adding new knowledge.
- [ ] Deductive inferences go beyond information we already have, thus they amplify our knowledge when they are used.
- [x] In valid deductive inferences, the conclusion must be true if the premises are true.

3: Match to get correct examples of a generalization, a direct inference and a projection.

The 36 novels examined so far all include a romantic couple.	Therefore, all novels include a romantic couple.
The 32 novels examined so far all have a female main character.	Therefore, the next novel we examine will have a female main character.
32 out of the 100 studied novels have a female main character.	Therefore, 32% of all novels have a female main character.

4: Below are some inductive and some deductive inferences. Which of them is the best example of a generalization, a projection, a Modus Ponens inference and a Modus Tollens inference respectively? Match each inference to the correct label!

If the train was late, then Jaya missed the meeting. Jaya didn't miss the meeting. Therefore, the train wasn't late.	Modus Tollens
The train has been late the previous 10 days. Therefore, the train will be late today.	Projection
The 500 examined samples of the new alloy X are all resistant to corrosion. Therefore, the new alloy X is resistant to corrosion.	Generalization
If it is raining, then the streets are wet. It is raining. Therefore, the streets are wet.	Modus Ponens
If I eat chocolate, then I am happy.	Not an inference

5: "Hume's problem of induction" is an argument saying that inductive inferences cannot be justified. What are its premises? Mark all correct options.

- [x] That a particular inductive inference rule cannot be justified using deduction.
- [ ] That inductive inferences are unscientific and should be avoided.
- [x] That a particular inference rule must be justified using induction and another inductive inference rule, which in turn also needs inductive justification and so on.
- [ ] That there are many kinds of inferences and that inductive arguments are justified by currently unknown inference types.
- [x] Inductive inference rules also require justification, and some kind of inference is needed to do this.
- [x] That every inference is either inductive or deductive.

6: Why is Hume's problem of induction problematic for the justification of inductive inferences, and thus an important challenge for science? Mark all correct options.

- [ ] Because if Hume is correct, all inductive inferences give necessarily true conclusions which means they are really deductive. Deductive inferences cannot be used in science.
- [x] Because if Hume is correct, the use of inductive inferences rules can only be justified by using other inductive inference rules, which leads to an infinite regress. Science uses a lot of induction and would therefore often not be justified.
- [ ] Because if Hume is correct, inductive inference rules would be unreliable and science would have to rely on deductive inference rules, which are often not as valid as inductive inference rules.

7: Mark all true claims about the Hypothetico-Deductive Method (HDM).

- [x] HDM is a method for testing hypotheses.
- [ ] HDM is based on deduction and not on induction. Therefore, it circumvents Hume's problem of induction.
- [x] HDM uses both deduction and induction.
- [ ] HDM is a scientific ideal that is rarely used in practice

8: Which of these are main components of the Hypothetico-Deductive Method (HDM)?

- [ ] If the observable consequences appear then there is some deductive support for the hypothesis being true.
- [ ] If the observable consequences appear then the hypothesis can be rejected using deduction.
- [x] If the observable consequences appear then there is some inductive support for the hypothesis being true.
- [x] Test if the observable consequences appear.
- [x] Deduce observable consequences from the hypothesis.
- [x] If the observable consequences do not appear then the hypothesis can be rejected using deduction.
- [x] Formulate a hypothesis.
- [ ] Formulate a research question.

9: Which of the following are requirements for a statement S to be used as a hypothesis?

- [ ] If S is to be counted as a good hypothesis, it should either be necessarily true or have the form of a question.
- [ ] S must be formulated in such a a way that it is easily understandable for laypersons.
- [x] S mustn't be necessarily true or necessarily false.
- [x] S must be on a form such that it can be true or false.
- [x] If S is to be counted as a good hypothesis, it should either be somewhat general or concern something that is not directly observable.

10: Falsification and falsifiability are not the same thing. Which of the following claims are true?

- [x] Falsification is an event, the observation that an implication of a hypothesis is not true, which entails that the hypothesis itself is false.
- [ ] Falsification is an event, the observation that an implication of a hypothesis is not true, which means that the test procedure went wrong (and should thus be modified).
- [ ] Falsifiability is a property of a hypothesis, it has to do with the number of observable consequences that actually have demonstrated that it is false.
- [x] Falsifiability is a property of a hypothesis, it has to do with the number of observable consequences that could show that it is false.

11: Mark all true claims about Popper’s falsificationism.

- [x] Falsificationism is in part an attempt to avoid Hume’s problem of induction.
- [x] According to falsificationism one can infer the falsity of a hypothesis from even a single false observable consequence.
- [ ] Falsificationism infers the truth of a hypothesis if all (so far) observable consequences have been observed.
- [x] Falsificationism is similar in its structure to the falsifying part of the hypothetico-deductive method.

12: What is the Duhem/Quine thesis? Mark all correct options.

- [ ] It is a thesis that is used by Popper as an argument against Hume's problem of induction.
- [ ] It is a thesis that states that hypotheses have to be tested using empirical equipment and cannot be conclusively falsified using theory.
- [x] It is a thesis that states that all hypotheses are tested using auxiliary hypotheses.
- [x] It is a thesis that introduces a problem to Popper's answer to Hume's problem of induction.
- [x] The thesis implies that no hypothesis can, strictly speaking, be conclusively falsified.

13: What is true regarding hypothesis confirmation? Mark all correct options.

- [ ] Degrees of confirmation can only be described quanitatively.
- [x] Frequentism states that one cannot meaningfully assign probabilities to hypotheses.
- [x] A problem for confirmation is under-determination: that there are many competing hypotheses that are compatible and supported by an empirical observation.
- [x] Degrees of confirmation can be described qualitatively or quantitatively.
- [ ] Frequentists argue that we should assign probabilities to hypotheses.

14: Which of the following scenarios are examples of choices that can be made to contravene the problem of under-determination? Mark all correct options.

- [x] The collected data on the population change over time in the area could be described by either a linear or a quadratic function. Both fit the data equally well, so we choose the linear function.
- [ ] If our hypothesis is true it will have three separate observable effects. However, none of them give deductive proof of the hypothesis, so we have to keep looking for observable effects that do.
- [x] If this crop has been affected by toxins in the ground, then the observable consequences will be discoloration and underdevelopment of the seeds. However, we know from previous studies that some common types of vermin cause the same kind of discoloration (and that there is no vermin causing underdevelopment). Hence, we concentrate our testing on underdevelopment of seeds.

15: What is true about the concept severe test? Mark all correct options.

- [x] When setting up a severe test for a hypothesis we exclude observable consequences that are irrelevant even if they are compatible with the hypothesis.
- [x] A problem with severe testing is that it requires a lot of background knowledge about possible alternative hypotheses.
- [x] Re-using observational data poses a problem to severe testing.
- [x] Severe tests should be relevant, in the sense that the observations we make would be very unlikely if the hypothesis were false.
- [ ] Curve-fitting is always considered a type of severe testing.

Quiz 2: Observations and measurements

1: What is empiricism? Mark the correct option.

- [ ] The thesis that pure reasoning is more reliable than observations.
- [x] The thesis that sensory experience is the ultimate basis for knowledge.
- [ ] The thesis that a statement is meaningless if it cannot be empirically verified.
- [ ] The thesis that all observations are theory-dependent.

2: For each observation below, decide whether it is (a) direct (unaided), (b) aided direct, or (c) indirect.

Observing a person's political views by letting them answer a survey	Indirect observation
Observing the the stellar constellation Ursa Minor by locating it in the night sky on the basis of its shape	Direct observation
Observing a person in a pitch dark room by using thermal imaging binoculars	Indirect observation
Observing the (approximate) temperature of a liquid by dipping one's finger into it	Direct observation
Observing a traffic sign from a distance while wearing contact lenses	Aided direct observation

3: What does it mean to say that an observation is theory-dependent? Mark all correct options.

- [ ] It means that there is only one theory that can account for the existence of the observational content (that which is observed).
- [x] It means that the justification and/or interpretation of the observation rests upon an assumption that one or more background theories are true.
- [ ] It means that, as of yet, the observation has not been made, although it's predicted by one or several scientific theories that are accepted as truth.

4: In the lecture, it is argued that logical empiricism is an unreasonable view on the relationship between observations and theory. What arguments are given in the lecture? Mark all correct options.

- [ ] Logical empiricism rests on the difference between observational sentences and theoretical sentences (theories), but this difference is a chimera. There is no such difference between sentences.
- [x] Indirect observations always require theory. Hence, there cannot be any observational sentences about indirectly observable properties that could act as a foundation for some theory.
- [x] Some aided-direct observations heavily rely on theory and are used to observe phenomena that cannot be observed by, and thus compared to, unaided-direct observations. Hence, some aided-direct observations are not acceptable according to logical empiricism, and this is counter-intuitive.
- [ ] Logical empiricism's idea of deriving theory from observational sentences is ultimately an idea based on the foundationalist view on justification. Hence, it cannot be correct because foundationalism is simply an incorrect view on the nature of justification.

5: In some cases, scientists choose to rely on indirect observations - even though the target phenomenon is, in principle, directly observable. What might their reasons be for doing so?

- [ ] Indirect observations are untouched by philosophical criticism - since they stem from methodological considerations that are widely used in science.
- [x] Direct observations are sometimes prone to errors that indirect observations are not prone to.
- [x] Direct observations typically only give qualitative properties.

6: What is the relationship between indirect observations and operationalizations?

- [x] An indirect observation is the act of observing something indirectly. An operationalization is a theoretical construction of an indirect observation (which might be put to work).
- [x] An operationalization is a causal or correlation hypothesis that must be true in order for a particular indirect observation to be accurate and precise.
- [ ] The terms "indirect observation" and "operationalization" are effectively synonymous.
- [ ] An operationalization is an indirect observation which is not theory-dependent.
- [ ] An operationalization is a technically advanced method of indirect observation that is applicable only for testing a very narrow range of scientific hypotheses.

7: True or false: It is possible that an operationalization might be sufficiently good (given one's purposes) even though the phenomenon in question might be operationalized in several other ways.

- [x] True
- [ ] False

8: Which of the following are quality criteria for operationalizations? Mark all correct options.

- [ ] The phenomenon of interest must in principle be impossible to observe directly.
- [x] The hypothesized relationship between the phenomenon of interest and the effect must be valid according to our best knowledge.
- [ ] The hypothesized relationship between the phenomenon of interest and the effect is theory-dependent.
- [x] The phenomenon of interest must be sufficiently well-defined.
- [x] The effect must be observable with sufficient precision.
- [x] The hypothesized relationship between the phenomenon of interest and the effect must be sufficiently stable for practical purposes.

9: Some arguments for and against operationalism are discussed in the video. Compare each of the following arguments with the arguments in the video: is it an argument for operationalism, against operationalism, or is it irrelevant? Match each argument with the correct description.

By defining a concept, e.g. happiness, as the operation by which it is observed/measured, e.g. the score one gives on a scale ranging from 0-5, you have effectively identified happiness as the particular number given. Since this number is directly observable, you now don't have to rely on any theory to observe happiness.	For operationalism
Operationalism is by consequence of its thesis an effective way to generate an operationalization. The connection between this operationalization's feature of interest and directly observable effect is automatically justified by means of the feature of interest's definition.	Irrelevant
By defining concepts through the operations by which they are observed/measured, you effectively split a concept into several subconcepts. Therefore, you create an abundance of unnecessary concepts which intuitively ought to be one and the same.	Against operationalism
A researcher who embraces operationalism and observes Alzheimer's disease via a memory test will have to say that Alzheimer's disease is the inability to generate a certain score on this memory test, when Alzheimer's disease in fact is something in the brain.	Against operationalism

10: Which of the following are examples that apply to one, two, or none of the two quality criteria for the specification of a unit of measurement?

One measure of a second would be to use the time it takes for a Cesium atom to oscillate, approximately, 9 billion times.	Stability and comparability
One measure of water sample's magnitude would be to use, as a unit, a small ice block with a predetermined magnitude.	None
One measure of length would be to use a predefined prototype of length as a unit, such as a seconds pendulum (a pendulum whose period is two seconds).	Comparability

11: Match each scale type with the description that uniquely applies to it.

Description A: This scale allows for quantitative comparisons of different steps between points on the scale. The Celsius temperature scale is a scale of this type. Also, on this type of scale the null does not necessarily correspond to a lack of the measured property, and the unit itself is arbitrarily defined in terms of this property.	Interval scale
Description B: By merely naming the instances in a set, without any mutual comparison, we've successfully placed them on a scale of this kind.	Nominal scale
Description C: This scale allows for quantitative comparisons of different steps between points on the scale. The null corresponds to a complete lack of the measured property. The Kelvin temperature scale (K) is a scale of this type. Not only is it possible to do quantified comparisons of the different intervals on the scale, but is also possible to say that 80K is in fact twice as hot as 40K. However, on scales of this type, the unit is arbitrarily defined in terms of the targeted property.	Ratio scale
Description D: This scale allows for quantitative comparisons of different steps between points on the scale. The null corresponds to a complete lack of the measured property. In addition, the unit of measurement corresponds to some naturally occurring unit or constituent of the targeted property. Probability measures are placed on this kind of scale.	Absolute scale
Description E: By placing objects on a scale of this kind, we have successfully arranged them hierarchically in accordance with the different extents to which they instantiate a certain property. However, this hierarchical arrangement is purely qualitative. For example, the scale TERRIBLE-BAD-NEUTRAL-GOOD-GREAT is a scale of this type. The scale does not allow us to say e.g. that the step from BAD to NEUTRAL is equal to (or twice as large as, or a third of) the step from NEUTRAL to GOOD.	Ordinal scale

12: Assume that you want to measure how many times more expensive a certain shoe model is compared to certain other shoe models. This requires a scale that orders your measurement and specifies an equivalent distance for each of the scale's steps. Given these considerations, the scale that you use is an interval scale.

- [ ] True
- [x] False

13: Measurement errors come in two types: random and systematic errors. What characterizes a random error? Mark all correct options.

- [x] Some random errors can be avoided by trying a different operationalization.
- [ ] Random errors do not affect the precision of the measurement.
- [x] Given sufficiently many measurements of a quantity and the absence of systematic errors, the average result of those measurements is expected to be precise.
- [ ] Random errors cannot be reduced by repeating measurements and averaging over outcomes.

14: What is true regarding systematic error? Mark all correct options.

- [x] Systematic errors affect the accuracy of a measurement.
- [x] Systematic errors might occur due to poor calibration.
- [x] A series of measurements might be very precise, even though they are affected by systematic error.
- [ ] Systematic errors affect the precision of a measurement.
- [ ] Systematic errors and random errors cannot occur at the same time.

15: How is it possible to get some indication of whether a method of measurement is suffering from a systematic error? Mark all correct options.

- [ ] We can test for precision.
- [x] We can test for convergent validity.
- [x] We can test for divergent validity
- [ ] We can check whether one can deductively prove that the theory underlying our operationalization is true.

Quiz 3: Experiments

1: Mill's method of difference involves setting up two separate systems S1 and S2. What is true about these systems? Mark all correct options.

- [x] Any background factors that are controlled in S1 must also be controlled in S2.
- [ ] If we do an intervention in S1, then we only need to observe the result in S1, not in S2.
- [ ] Any intervention done on S1 must also be done on S2.
- [x] Using two systems enables us to exclude alternative causes of the observed results.

2: Which of the following statements about Mill's Method of Difference are true?

- [x] Mill's Method of Difference exhibits the reasoning behind testing hypotheses with experiments.
- [x] The main question that Mill's Method of difference asks is: what causes event E?
- [x] Mill's Method of Difference elucidates the ways in which an experimental process can go wrong.
- [ ] Although it is an illustrative thought experiment, Mill's Method of Difference has little to do with how actual experiments are conducted.

3: Which of the following empirical practices constitute experiments? Mark all correct options.

- [ ] Observing the effects of a natural catastrophe on a local economy.
- [ ] Tracking the orbits of two of Saturn's moons to see if they're affected by each other's gravitational pull.
- [x] Testing a medical treatment for potential side-effects by giving it to a number of patients and observing their health.
- [ ] Measuring the temperature of arctic lakes by taking samples at different depths.
- [x] Measuring the boiling point of a liquid in a pressure chamber to observe how this changes when you change the pressure.

4: A team of researchers want to test if a new nanomaterial for jackets that they are developing is more dirt repellent than an ordinary textile material for jackets. For this they plan to conduct a field experiment. What relevant actions for this experiment could they take? Match each action below to Control, Intervention, Observation of the outcome, or Irrelevant or not directly relevant.

Measuring the amount of dirt on the jackets.	Observation of the outcome
Estimating the average age, weight and food intake of the participants during the trials.	Irrelevant or not directly relevant
Letting participants wear jackets made of the nanomaterial.	Intervention
Making sure that the participants in the experiment are exposed to similar environments and amounts of dirt, and that the trials are conducted for an equal duration of time.	Control
Making sure no participants wear any smartphones or communicate with their families or friends during the trial.	Irrelevant or not directly relevant

5: Match the correct description with the correct observational practice.

An observational study where circumstances are organized as if there had been manipulation for intervention and control, but no manipulation is in fact done.	Natural experiment
An experiment where there is manipulation in terms of an intervention but where several background variables are not controlled, in particular not controlled through manipulation.	Field experiment
A model study where the model is created as to mimic an experiment, but where the variable that is intervened on is only represented rather than being actually present.	Simulation experiment
A study without intervention, where some background variables might be (but need not be) controlled.	Observational study

6: Imagine the following scenario.

You are the head researcher at AI Robotics Inc. In six months' time you are going to release a new line of humanoid robots. You want to know how well they are going to keep their balance doing regular household tasks such as picking up the newspaper, cleaning up, and similar things. To get some data to answer your question, you build a computer program that constructs a virtual 3D environment in the form of a house, and impose artificial gravity onto the environment. You then import the schematics for the new humanoid robot into this environment and then watch it perform such household tasks in the virtual world.

Did you just perform an experiment? Mark all correct options.

- [ ] No. We constructed an observational study.
- [x] No. Simulations are not experiments because they do not manipulate the real variables.
- [ ] Yes. Experiments are mostly about predicting what will happen, and this simulation gives us information to base our judgement on.
- [ ] Yes. We controlled the background variables, manipulated our target variable and observed or recorded the outcome.

7: Which of the following, broadly construed alternatives describe general types of error in experiments? Mark all correct options.

- [ ] A failure to reconcile the hypothesis with the common sense conception of the investigated phenomenon.
- [x] A failure to implement control over background variables.
- [x] A failure to correctly observe the outcome of the experiment.
- [x] A failure to intervene on the target

8: Match each term with the correct description.

A factor influences the selection of the sample or the division into test and control group, creating some effect which one did not aim to create.	Selection bias
An inference about an experimental system is internally valid if there is a relation between intervention and observed effect inferred from the experiments is indeed true and is not confounded by uncontrolled background factors.	Internal validity
The observation is registered incorrectly due to psychological properties of the observer.	Confirmation bias (interpretation problem)
The act of observation changes that which is observed.	Observer effect (influence problem)

9: Imagine that you are performing an experiment…

Imagine that you are performing an experiment on the heat emission of a certain type of exhaust engine. Your hypothesis is that a new kind of material used in the engine will result in a more even heat emission. You set up a prototype of the engine in a laboratory environment and carefully plan every step of the experimental procedure, including what equipment you will use to control background factors and measure the relevant parameters. You then perform the experiment according to your plan and note the results, which support your hypothesis.

What does it take for your experiment to be repeatable? Mark all correct options.

- [x] The results as well as the mentioned details about experimental setup, measuring equipment etc. must be published or made available in some way, so that other researchers can take part of the information.
- [ ] None of the other alternatives are required. If all steps specified in the question text are taken then, by definition, the experiment is repeatable.
- [ ] A separate research team must test the same hypothesis and obtain the same result.
- [ ] Your own laboratory and equipment must be made available to future researchers.

10: Imagine the following scenario.

You have been tasked to investigate whether Bright Light Therapy affects people's self-assessments of their level of well-being. You decide to conduct an experiment as a way of providing an answer to this question. A sample is randomly generated from a population of people who've agreed to participate in the study. You are now to decide which control measures to implement in your investigation.

Which of the following options are relevant measures for establishing control?

- [x] You expose only a part of the group to the bright light therapy and only pretend to do so for the remaining people.
- [x] You repeat the experiment in different times of the year.
- [x] You make sure that the population from which the sample is taken is not skewed in some sense.
- [ ] You identify the participants randomly chosen and select only the ones you think will lead to the experimental result you are expecting.
- [ ] You influence the assessments people make of their level of well-being when answering a standardized questionnaire.

11: Below are examples of different kinds of blinding. Match each example with the most relevant type of blinding.

An external laboratory collected the material samples from both the test and control group and gave each sample a random label before returning them to the researchers.	Experimenter blinding
When designing the survey, we included questions about a wide range of subjects without indicating that the study was only about eating habits.	Single (subject) blinding
The researcher was given two batches of visually identical pills and then administered a randomly chosen pill to each test subject, carefully noting which batch the pill came from.	Double blinding

12: In some experimental cases, it's not possible to eliminate all background factors that (might) have a disturbing effect on the outcome. In such a situation, is it necessarily the case that using a model simulation is a preferrable method of investigation? Mark correct option.

- [x] No. Disturbing factors cannot always be eliminated. While a preferred control measure in many instances, sometimes it suffices to register different effects separately and take them into account in the analysis of the data.
- [ ] Yes. Models can provide us with the data required to answer questions better than an experiment can in almost all instances.
- [ ] Yes. Disturbing factors always influence tests in ways that must be controlled for through elimination.
- [ ] No. It is always possible for us to, in some more or less abstract way, eliminate disturbing background factors. Therefore using a model is not a preferred method of investigation.

13: Imagine the following scenario.

You want to test how sensitive a set of plants are to a certain chemical compound. First, you devise a way of safely subjecting such plants to this compound. Second, you prepare yourself to record the levels of certain toxins in the plants throughout a period of two weeks. Before all of this, you wish to separate the plants into a treatment and a control group.

Which methods below can be used for dividing the plants into control and treatment group? Mark all correct options.

- [ ] Use a lottery wheel to distribute the items based on whether they land on a red or blue area (of which there are equally many) on the wheel, and repeating this process until known disturbing factors only occur in the control group.
- [x] Flipping a fair coin and distributing the groups based on which side it landed, and check that known relevant factors are indeed equally distributed in treatment and control group.
- [x] Use a random number generator and distribute items to the two groups based on whether the number was even or odd, and check that known relevant factors are indeed equally distributed in treatment and control group.
- [ ] Forming a treatment group of the plants that seem healthiest according to some known parameters and putting the rest in the control group.

14: What are some purposes for experimenters' choice of using the method of randomization?

- [x] Randomization facilitates blinding.
- [ ] Randomization helps to establish the selection based on one singular factor.
- [ ] Randomization ensures that all background factors are evenly distributed throughout the groups.
- [x] Randomization eliminates the bias of consciously or unconsciously choosing particular items from the sample.
- [x] Randomization helps convince others that you have not rigged the treatment/control division in favor of the outcome you want.

15: What is true about external validity? Mark all correct options.

- [x] Randomly choosing a sufficiently large sample from the population is a way to ensure external validity.
- [x] To ensure external validity we need our sample to be representative of the population.
- [ ] Randomly assigning our sample into control and treatment group is a way to ensure external validity.
- [ ] The results of an RCT (randomized controlled trial) will by design be externally valid.

Quiz 4: Models

1: What are possible reasons why researchers/scientists use representations instead of investigating targets directly? Mark all correct options.

- [ ] Some targets are not scientifically interesting; we already understand them.
- [ ] Some targets are easily understood; we often want to see something in a more complex way.
- [x] Some targets are not physically accessible; they are, for example, too far away.
- [x] Some targets are too complex to investigate directly; we cannot represent all aspects of the target.

2: If we find out that a theory is false, then we cannot use the theory as a model. True or false?

- [ ] True
- [x] False

3: Consider the following scenario. A person wishes to model the flow of water through a floodgate. She can either choose to model the flow of water using a simplified computer simulation, or she can construct a scale model of the floodgate in question. Which claims are true?

- [ ] She can only use the scale model, because simplified simulations are not sufficently customized for real world targets.
- [ ] She can use either of the two models. Which one she chooses should primarily depend on the transparency of the model.
- [x] She can use either of the two models. Which one she chooses should primarily depend on the purpose of the model.
- [ ] She can only use the computer simulation, because constructions are inherently unreliable.

4: Assume that you are using a rat as a model for a human - but rats and humans are different in many ways, and similar in others. Familiarize yourself with the terms negative, neutral and positive analogies by matching each description with the correct term.

Brain size being different	Negative analogy
General organ physiology is similar	Positive analogy
Hormones are the same	Positive analogy
A drug reduces obesity in rats, but has not been tested on humans.	Neutral analogy
Lifespans are different	Negative analogy
Properties we have identified in the rat, but that we don't know about in humans (yet)	Neutral analogy

5: Decide which of these alternatives are similarities and which are differences between models and experiments. Match the alternatives with the correct description.

How certain we are that the result stems from our assumptions and not something else is a...	Difference
That we observe results in both models and experiments is a...	Similarity
The need for justifying the assumed similarity between what you are studying and what you want to find out about is a...	Difference
Being able to manipulate factors is a...	Similarity
The source of the errors is a...	Difference

6: Is similarity itself a recipe for good representations? Mark all correct options.

- [ ] Yes. Necessarily, a model with a higher degree of similiarity is a better representation than a model with a lower degree of similarity.
- [x] No. Similarity is only important with regard to purpose-relevant properties.
- [ ] Yes. All models must be similar in all properties.
- [ ] No. Similarity in itself is not important, only the purpose of the model is.

7: Imagine that you are to model the growth and decline of a population over time using a faucet to fill a transparent bucket of water with marked lines for the volume of liquid in it. Would this model be similar enough to serve its purpose? Mark all correct options.

- [ ] No. This model is not similar to the target in any respect.
- [ ] Yes. Growth is about the addition of people which the added water can represent fully.
- [x] No. This does not allow for populations decreasing, perhaps a plugged hole should be added to the bucket.

8: Assume that a model M1's parameters are implied by model M2's parameters, Which model is more precise? Mark the correct option.

- [ ] M1 is more precise.
- [x] M2 is more precise.

9: Assume that the purpose is to use two globes as representations of the relative size of the moon and the sun, which of the following models are the simplest ones?

- [ ] A scale model of the moon and a scale model of the sun, that both take into account their position in the planetary system and their respective orbits.
- [x] A scale model of the moon and a scale model of the sun, both made of plastic.
- [ ] A scale model of the moon and a scale model that introduces the span of solar flares.
- [x] A scale model of the moon and a scale model of the sun, both made of paper.

10: Robustness is a desirable virtue of models. Which of the models described below are robust?

- [ ] A model whose second (but not first) condition is changed and yields a different result.
- [ ] A model which can have none of its parameters changed.
- [ ] A model which replaces none of its parameters but yields different results when applied to the same data.
- [x] A model which has some of its conditions changed but yields the same result.

11: Match each model example with its most prominent model virtue. Note that some of these models might have other virtues as well. We are simply asking what virtue would seem most prominent based on the short description of the model (and potentially other information).

In order to get a better understanding of the danger in real car crashes, researchers have modelled car crashes with real cars and crash dummies. The dummies' properties (e.g. their spines) are supposed to be, as much as possible, identical to the same properties in real humans.	Similarity
For any attempt at modelling a language that is complete and sound (with sufficient expressive power), it will be possible to deductively show that there will exist unprovable sentences in the language. This result is known as Gödel's (first) incompleteness theorem.	Analytical tractability
Viewed as a model, Newton's laws of motion could – at the time – explain every observed object with fewer principles, as compared to the older Aristotelian view of motion and force.	Simplicity
In economics, an individual's demand for public goods U can be modelled according to a utility function F(X, G) where X is the individual's private goods and G is the available public goods. The relation between X and G can be described via relatively simple mathematical equations (which most people can follow and understand).	Transparency
Statistical outliers are often problematic when doing a regression analysis whose underlying assumptions are false, in a given case. However, certain regression-analysis models are designed to be able to handle such outliers when the underlying assumptions (e.g. about a variable's normal distribution) indeed are false.	Robustness

12: In the lecture, some reasons are brought up for using mirror-modelling rather than other modelling strategies. What are these? (Disregarding for the moment possible counterarguments.)

- [ ] What makes mirror-models ideal is the high degree of idealization, as compared to other modelling strategies.
- [x] If the model mirrors its target closely enough, then performing tests on the model will give us results that are as justified as if we had performed an experiment on the real target.
- [ ] By picking out a few features and modelling them in isolation the mirror-model gives a clear answer, even if the mirror-model and the target are quite different.
- [x] Since the model mirrors so many aspects of its target, we might actually be able to generate predictions concerning the target.

13: Assume someone says: "When modelling the train car for the purpose of studying its behaviour when colliding with a car at an intersection, I am looking for a very life-like model where some minor change(s) to the car's weight distribution would not dramatically alter the result." Mark all correct options. You suggest that he/she should create a computer mirror-model because…

- [x] A mirror model is often precise. That is, the specifications of a mirror model, which are highly accurate to real collisions between trains and cars, will imply the specifications of a model with less accurate specifications.
- [ ] A mirror model is often transparent. That is, (assuming it's a good mirror model) due to the model's very high similarity, it will be possible for someone using the mirror model to grasp why a particular result is given as an output.
- [ ] A mirror model is often simple. That is, it will be relatively few properties to take into account when creating this model in order to make it mirror real collision between cars and trains.
- [x] A mirror model often has the similarity that is relevant for the purpose. That is, the mirror model will, if created with great care, be realistic to real collisions between cars and trains.

14: What is a requirement when using isolating models? Mark all correct options.

- [ ] That the model does not represent the real world target.
- [ ] That no other kind of model is a practical alternative.
- [ ] That what you are studying does not consist of too many parts.
- [x] That what you are studying can be divided into different parts.

15: What is the relationship between the modelling virtues "similarity" and "simplicity"?

- [x] When evaluating two models, their level of similarity is assessed by comparing them to their target and their simplicity is evaluated by comparing them to each other.
- [ ] Simplicity and similarity are opposites. That is, when a model's similarity goes up, its simplicity goes down.
- [ ] One can determine if a model is simple just by counting the number of model assumptions or model components.
- [x] A model may be both simpler and more similar to the target in comparison with another model.
- [x] No model is simple or similar just in itself; they always need to be compared to something else.

Quiz 5: Statistics

1: According to the definition of "lying with statistics" as defined (quite narrow) in this course, which of the following problematic practices are likely to be regarded as cases of lying with statistics?

- [ ] Using inaccurate data to draw a true conclusion.
- [x] Formulating the conclusion of a statistical study using a term such as "average", without specifying, for instance, whether the median or mean is intended.
- [ ] Making a claim for which there is absolutely no statistical basis.
- [x] Using accurate data and valid statistical methods to draw a false conclusion.

2: Which of the following claims about the choice and justification of statistical methods are true? Mark all correct options.

- [ ] The available statistical methods will all yield the same result if applied to the same data.
- [ ] The use of a certain statistical method is a conventional matter, and the choice of this method can be left to a statistical program.
- [x] The justification of statistical methods is an integral part of science.
- [ ] The choice of statistical methods does not matter if you know which conclusion you need to arrive at.

3: Imagine that you are a researcher…

Imagine that you are a researcher in psychology who is interested in developing a statistical test of the positive impact on one's ability to perform well on standardized tests, by being exposed to so-called “intelligence-related words” (for example “smart”, “clever”, “genius” and so forth). To determine this, you measure people’s performance on such standardized tests before and after having been exposed to intelligence-related words. You note that 75 % of those who were subjected to intelligence-related words improved their score the second time they took the test.

Is it true or false that one can conclude this result to be statistically significant solely on the basis of this information?

- [ ] True
- [x] False

4: In the example from the lecture, the conclusions derived from the data set of the European Central Bank allowed for two opposing conclusions; namely that Germany was the worst off in terms of household income on average, and that they were, on average, somewhere in the middle. How is such a seeming contradiction possible? Mark the correct option.

- [x] It is possible because ‘average’ is ambiguous, and two different senses were used.
- [ ] It is in fact not possible. People simply did not understand the data.
- [ ] It is possible because there is no truth of the matter.

5: Which of the following are examples of good statistical methodology?

- [ ] Using any of the methods that are implemented in some high-quality statistics software.
- [ ] Carefully choosing the statistical method that is the most suitable one to get a significant result, given the type of study you are performing.
- [x] Carefully choosing the statistical method that is the most suitable one to get a true result, taking into consideration the type of study you are performing.

6: Which of the following situations might occur depending on how we interpret 'average' in the sentence "this student's test score is below the average score"? Mark all correct options.

- [x] If there are some high-scoring outliers (and no low-scoring outliers), then the student's test score might be below the mean score but still be above the median score.
- [ ] If there are some high-scoring outliers (and no low-scoring outliers), then the student's test score might be below the median score but still be above the mean score.
- [x] If there are some low-scoring outliers (and no high-scoring outliers), then the student's test score might be above the mean score but still be below the median score.
- [ ] If there are some low-scoring outliers (and no high-scoring outliers), then the student's test score might be above the median score but still be below the mean score.

7: What are some common sources of misunderstanding connected to the choice of statistical format? Mark all correct options.

- [x] A relative risk reduction may be overestimated when compared to the same risk reduction in absolute numbers.
- [x] It may be hard to understand how absolute risk reduction is connected to the "number needed for treatment" format.
- [x] A relative risk increase of e.g. 200% may be perceived as alarmingly large if one doesn't understand that the initial risk was very low.

8: Under what circumstances do we have reason to evaluate a hypothesis statistically?

- [x] In order to account for measurement errors and similar disturbing influences, one might want to determine the acceptable error of a test. Quantifying this error requires statistical methods.
- [x] For some hypotheses, the only relevant implications are stochastic. The truth of these implications can be determined precisely only with the help of statistical tools.
- [ ] We should always do so. Otherwise, the testing procedure is unscientific.
- [ ] When the hypothesis is stated in terms of statistical notions, such as 'mean' and 'median', we should always apply statistics.
- [x] In order to correctly determine how much a set of observational data changes one's confidence in a hypothesis, statistical methods are useful.

9: Which of the following claims about p-values are true (in the context of Fisher's version of a significance test)? Mark all correct options.

- [ ] Given a hypothesis H (with a test statistic T) and a data set D, the p-value is the probablity of H given D. That is, it is the probability that one's hypothesis is true, given the outcome of one's test.
- [x] P-values are compared with a significance level
- [ ] P-values are set by convention.
- [x] Given a hypothesis H (with a test statistic T) and a data set D, the p-value is the probablity of D (or any more extreme outcome) given H. That is, it is the probability of getting the outcome one actually got, or a more extreme one, given that one's hypothesis is true.

10: Imagine a researcher…

Imagine a researcher who is investigating whether changing the materials used to conduct electricity in central processing units (CPUs) positively affects its efficiency by a factor of 1.12. The researcher formulates the hypothesis "the effect of changing the materials is less than 1.12". When failing to reject this hypothesis, the researcher decides (against good research practice) to perform p-value abuse.

Which of the following alternatives constitute such abuse? Mark all correct options.

- [x] Expanding the sample in order to find a data set that (in isolation) has a significantly low p-value in relation to the stated hypothesis, and then choose to report only that data set.
- [ ] Rejecting the stated hypothesis without reporting the p-value of the data.
- [x] Using the same data sample and, with the only reason being to generate a statistically significant result, modifying the value in the hypothesis to 1.15, which gives a significant p-value.
- [x] Purposely choosing a significance level that is higher than the p-value calculated for the test data.

11: Neyman-Pearson's method of hypothesis testing allows researchers to choose between two competing hypotheses. What is true about this method? Mark all correct options.

- [x] Committing a type I error in a test excludes the possibility of a committing a type II error at the same time.
- [x] The hypotheses must be mutually exclusive and jointly exhaustive.
- [ ] Usually, one of the hypotheses is taken to be the main hypothesis. The other can be just about any given hypothesis, provided that it functions as an auxiliary hypothesis in some possible empirical test of the first hypothesis.
- [x] Rejecting one hypothesis entails the acceptance of another.

12: Suppose you have set up a Neyman-Pearson test with the hypothesis Hi: “the new advertising campaign doesn’t increase the sales of the advertised products”. What is true about the power of the test? Mark all correct options.

- [x] Collecting data from more people in the area targeted by the campaign is one way to increase the power of the test.
- [x] Collecting data from more people in the area targeted by the campaign will make your test less probable to reject Hi in situations when Hi is in fact true.
- [x] Collecting data from more people in the area targeted by the campaign will make your test less probable to accept Hi in situations when Hi is in fact false.
- [x] If your test is high-powered and the campaign does in fact increase the sales, then there is a high probability that Hi will be rejected by the test.
- [ ] If your test is high-powered and the campaign does in fact increase the sales, then there is a high probability that Hi will be accepted by the test.

13: In a sentence, what is the focus of Bayesian statistics? Mark the correct option.

- [x] Calculating the probability of the hypothesis being true, given the evidence.
- [ ] Calculating the probability of observing the evidence, given that the hypothesis is true.

14: Three different problems with Bayesian statistics were mentioned in the lecture: (1) the problem of determining priors, (2) the problem of old evidence, and (3) the problem of uncertain evidence.

Sometimes, it is hard to distinguish the evidence used in determining the prior probability of a hypothesis from the evidence used in a Bayesian evaluation of the probability of that hypothesis.	The problem of old evidence
Bayesian statistics require that we quantify prior probabilities, yet does not specify how such probabilities should be determined. Thus, people with radically different priors might not be able to come to an agreement on whether a hypothesis is more likely to be true or more likely to be false - even if they have access to the same data.	The problem of determining priors
Prior probabilities should not be determined on the basis of old evidence - yet, we often have nothing else to go on.	None of the three problems
Bayesian statistics typically assumes that the probability of the evidence is 1. Yet, in reality, we are rarely (if ever) justified in being so certain about the strength of our evidence.	The problem of uncertain evidence

15: What is true about the differences between Fisher's significance testing, Neyman-Pearson hypothesis testing and Bayesian hypothesis testing? Mark all correct options.

- [x] Bayesian hypothesis testing can mirror the idea that humans have prior beliefs about the truth and falsity of hypotheses. This cannot be done with Fisher and Neyman-Pearson.
- [ ] Neyman-Pearson testing is the only one out of the three where the test can reject a hypothesis.
- [ ] As opposed to a Neyman-Pearson test, the hypotheses in a Bayesian test need not be mutually exclusive.
- [x] Neyman-Pearson hypothesis testing avoids some kinds of p-value abuse that Fisher significance testing is threatened by.
- [ ] Fisher tests and Neyman-Pearson tests include two hypotheses that are tested simultaneously, but Bayesian tests don't.

Quiz 6: Explanations and causes

1: Fill in the blanks.

According to the Deductive-Nomological account of explanation, the occurrence of a phenomenon can be satisfactorily explained by demonstrating that [*it was expected to occur***], by showing that [***it follows deductively from***] [***laws, or law-like generalizations***] and [***circumstances, or initial conditions***].**

2: An argument can be made that explanations should be asymmetric. If so, the Deductive-Nomological account of explanation is in trouble, since it is structurally symmetric. What does this imply? Mark all correct options.

- [x] It implies that there are cases that satisfy the conditions of the Deductive-Nomological account, but intuitively do not constitute genuine explanations.
- [ ] It implies that there are cases that fail to satisfy the conditions of the Deductive-Nomological account, even though they intuitively seem to constitute genuine explanations.
- [x] It implies that, on the account of the DN-model, it is sometimes possible to explain a phenomenon by pointing to its effects (in conjunction with relevant law-like generalities), rather than to its causes. This is quite counter-intuitive.

3: Use the options below to fill in premises and conclusion in order to get a sound but, in an obvious way, 'causally irrelevant' deductive inference, as discussed in the lecture. (Please assume that the options for premise 2 are all true).

Premise 1 (general claim)	Every billionaire that goes to the gym is wealthy
Premise 2 (initial condition)	Jane is a billionaire that goes to gym
Conclusion (explanandum)	Therefore, Jane is wealthy

4: What is true about the problem of causal irrelevance? Mark all correct options.

- [ ] If a DN explanation is causally irrelevant, then the deduction in the explanation is not valid.
- [x] If an explanation is causally irrelevant, then its explanans doesn't point out the difference-making causes.
- [x] The problem of causal irrelevance shows that certain accounts of explanation are not sufficient for all situations.
- [ ] The problem of causal irrelevance shows that certain accounts of explanation are not necessary for all situations.

5: Besides avoiding the problems outlined in the previous questions, what advantages do causal explanations have over Deductive-Nomological explanations? Mark the correct option.

- [ ] Causal explanations are always easier to understand.
- [x] Causal explanations, but not DN-explanations, can provide us with understanding of why a phenomenon occurred without making reference to any general, law-like regularities.
- [ ] Causal explanations, but not DN-explanations, can provide us with an understanding of why a singular phenomenon occurred.

6: Can you separate the different kinds of explanations? Some of the explanations below are DN explanations, while others are causal explanations but not DN explanations. Match each explanation below with the right type!

Why does particle P have amount x of energy? The energy of a particle is equivalent to the product of its mass and the speed of light squared (C^2). P has mass y. x = y * C^2. Therefore, the energy of P is x.	DN explanation
Why did Mr Jones burn his hand? Water vaporizes at 100 °C. Water vapor (above 100 °C) causes burn injuries to unprotected human skin. Water vapor rises upwards in normal conditions. The water in the pot was boiling at 100 °C. Mr Jones held his unprotected hand over the pot. Therefore, Mr Jones burnt his hand.	DN explanation
Why did the rock roll into the lake rather than stopping? The rock had a round shape, the hill beside the lake was steep, the rock was pushed down the hill, and there were no obstructions on the rock's trajectory. Therefore, the rock rolled into the lake.	Causal, not DN explanation
Why did Mr Jones get away unharmed instead of burning his hand on the water vapor? Mr Jones saw that the water was boiling and put on protective gloves. Therefore, when Mr Jones held his hand over the boiling water the gloves protected his skin. Therefore, Mr Jones got away unharmed.	Causal, not DN explanation

7: Is it problematic if an explanandum isn't contrastive? Mark all correct options.

- [x] Yes, because then the explanans might not provide the relevant kind of contrast.
- [ ] No, it is sufficient that the explanans is contrastive. Contrast in the explanandum is superfluous (and therefore often omitted).
- [ ] Yes, because explanations that aren't contrastive are false.

8: Each question below refers to an explanandum, and each answer to an explanans. Your task is to fill in the contrastive information in the question (explanandum), such that the answer (explanans) makes better sense.

Why did the vase break into large pieces, [*rather than the bowl breaking into pieces***]? Because the vase is made of a more fragile material than the bowl.**
Why did the dinosaurs go extinct, [*but not the mammals***]? Probably because the asteroid impact rapidly made the global climate unfit for the survival of some organisms, but not so much for the survival of other organisms.**

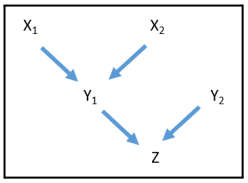
9: Which description fits best for each of the five quality criteria of causal explanations?

The explanans must identify all the contributing causes that produced the difference asked for in the explanandum.	Precision (of the explanans)
Is whether the explanans correctly describes the actual state or properties of the explanandum in the world.	Accuracy
The more precise the contrast is stated in the explanandum, the better the explanation is.	Precision (of the explanandum)
The less sensitive an accurate difference-making explanans is to background causes, the more powerful the explanation is.	Non-sensitivity (of the explanans)
The more easily a given explanation can be grasped, the more powerful it is.	Cognitive salience

10: Match each explanation with its most prominent explanatory virtue(s). Match one explanation to each right-hand option.

Why do trees exist? Because God created them.	Cognitively salient (but inaccurate)
Why does KTH have a course in calculus in several variables, when Karolinska Institutet does not have such a course? Because students at Karolinska Institutet are not interested in such a course.	Precision (in the explanandum)
Why was the train late last Sunday, when the train usually is on time? Because there was a sudden need for maintenance of the train rail, and all trains had to stop and wait for the maintenance to be completed. Maintenances are rare on this train-rail track.	Precision (in the explanans and explanandum)
Why does Sweden have a fundamental law called the freedom of the press regulation (tryckfrihetsförordningen)? Because the law was instituted by the Swedish parliament back in 1949 as a reaction on the harsh restrictions that governed media during the Second World War.	Accuracy and non-sensitivity

11: What is correct about the following graph of a causal scenario? Mark all correct options.



- [x] If an intervention is done on Y2, keeping all other Xs and Ys fixed, then Z will change.
- [x] The graph could be describing the following scenario: "If the knob is turned (X1) and the stove is connected to electricity (X2), then the stove heats up (Y1); and if also the pan of soup is on the stove (Y2), then the soup is cooked (Z).
- [x] X1 is a contributing cause of Z.
- [ ] If an intervention is done on X1, keeping all other Xs and Ys fixed, then Z will change.
- [ ] X2 is a contributing cause of X1.
- [ ] The graph could be describing the following scenario: "If the knob is turned (X1) and the pan of soup is on the stove (Y1), then the stove is connected to electricity (X2); and if also the stove heats up (Z), then the soup is cooked (Y2).

12: What is true regarding causation? Mark all correct options.

- [ ] Correlation in a data set is necessary for causation.
- [x] Causation measures the productive influence of one variable on another.
- [x] Causation is an asymmetric relation.
- [ ] Correlation is sufficient for proving causation.

13: What is true regarding correlation? Mark all correct options.

- [x] Correlation measures the association between two variables.
- [ ] Correlation measures the productive influence of one variable on another.
- [x] Correlation can be a sign of causation.
- [x] Correlation is a symmetric relation.
- [x] Correlation is not sufficient for proving causation.

14: Which of these are examples of causal models (causal scenarios) that are compatible with the observation that ice cream sales and the rate of murder in a population are correlated in a data set? (Note: they should be theoretically compatible, but you don't have to find them plausible.)

- [x] Eating ice cream makes a person more likely to kill another person.
- [x] Both ice cream and murder rates increase when the summer comes because people like to eat cold food when it is hot, and people are out and about more in the summer, making them easier to kill.
- [x] When someone has been murdered, people tend to console themselves by eating ice cream.
- [ ] When someone has been murdered, this increases the likelihood of revenge murders, increasing the total amount of murders. However, when someone has eaten a lot of ice cream, they become less interested in eating an additional ice cream.
- [ ] The observation is an artefact of the study and there is no causal relationship at all between the events, nor is there any other common cause.

15: Suppose that we have found that X and Y are correlated. Now we want to find out whether X causes Y. Which of the following is true? Mark all correct options.

- [ ] Instrumental variable analysis can only be used to show a correlation between X and Y, not a causal relation.
- [x] Mill's method of difference can be used to investigate whether X causes Y.
- [ ] The only way to prove that X causes Y is by performing some kind of experiment where we intervene on X.
- [x] Mill's method of difference can be used to exclude that X and Y have a common cause.

Quiz 7: Qualitative methods

1: Design often requires studying human behavior. Why is that? Mark all correct options.

- [ ] A designer might want to find out some previously hidden causal relationship in the artefact itself in order to achieve a better understanding of the artefact.
- [ ] A designer might be interested in the artefact's material, and whether the artefact could be constructed from some other material which has a certain virtue.
- [x] A designer might want to perform a design validation to check whether the artefact fulfils some functional requirements.
- [x] A designer might be interested in what particular functions an artefact should satisfy by studying potential users' demands.

2: In which ways do social science methods differ from natural science methods?

- [x] Interpretation of data typically involves attributions of mental states.
- [x] The focus is on human behaviour.
- [x] They produce qualitative data.
- [ ] There are no laboratory experiments.
- [ ] Validity of results is of no concern.

3: What does a belief-desire explanation consist in? Mark all correct options.

- [ ] The explanandum is deductively inferred from an individual's beliefs and desires.
- [x] The explanans includes beliefs and desires of an individual which together inductively infer the action to be explained - the explanandum.
- [x] The explanandum is explained through an explanans which refers to an individual's beliefs and desires.
- [ ] A belief-desire explanation of an individual's action explains why this individual acted the way she did only by reference to her desires.

4: What do you do when you adopt the intentional stance? Mark all correct options.

- [x] You eventually come to a conclusion about what the object ought to do relative the beliefs and desires you have ascribed to the object.
- [x] You try to understand what the object’s goal(s) is in order to understand how the object believes that it will reach this goal.
- [x] You ascribe a sort of rationality to the object – that is, rationality in the sense that the object has beliefs about how to reach its goals.
- [x] You attribute mental states to the object that you adopt the intentional stance towards.

5: Which of the following are correct ways to understand what the methodological problem of interpretation is about? In other words, what is at stake according to this problem?

- [x] The methodological problem of interpretation points out that even if we have an idea of how to model behavior, we still need a further refinement of what distinguishes good modelling from bad modelling, in a given context.
- [x] The methodological problem of interpretation points out that a given interpretation of behavior could, more or less, be arbitrarily chosen.
- [ ] The methodological problem of interpretation points out that the explanation model for behavior, i.e. the belief-desire model, is a faulty model.
- [ ] The methodological problem of interpretation points out that behavior can only be grounded in one desire and one belief.

6: One cannot do social science without interpretation.

- [ ] True
- [x] False

7: Which of the following are ways to avoid the the methodological problem of interpretation?

- [x] If it is possible then we could search for observable effects - via different operationalizations - that provide further evidence for the attribution of a particular mental state.
- [ ] We can avoid investigating at the level of social properties and instead focus on individual properties.
- [x] If it is possible then we could reduce the range of the interpretation.
- [x] If it is possible and reasonable, then one could investigate stimulus-response relations instead.

8: How can one ensure repeatability of a qualitative observation? Mark the correct option.

- [x] By making sure that the interpretation methods used are intersubjective, i.e. that the interpreting methods are transparent.
- [ ] By stating that if another researcher would use the same interpreting methods, then the same outcome would be observed.
- [ ] By making sure that if one were to use a different operationalization of the feature of interest, then one would get sufficiently similar outcomes.

9: Qualitative and quantitative data types have fundamentally different content.

- [ ] True
- [x] False

10: The following are examples of operationalizations using a survey, making a qualitative observation of the emotional attitudes of the subjects. Mark all options that would be problematic for such a qualitative observation.

- [x] The constructor of the questionnaire unjustifiably assumes that the expressions used in the questions are understood in the same way by everyone answering the questionnaire.
- [x] The subjects' emotional attitudes are represented on a measurement scale which fails to meaningfully convey the magnitude of the subject's real attitude (such as an absolute scale).
- [ ] The questionnaire does not yield a measurement of the subjects' attitudes that can be represented on either an interval or ratio scale.
- [ ] The researcher uses a lexical definition of the particular emotional attitude which is intended to be measured by the questionnaire.

11: Which of the following are examples of how one can increase the robustness of a qualitative observation? Mark all correct options.

- [ ] By creating several operationalizations of different properties, clearly aiming for divergent validity.
- [x] By creating several operationalizations of the property, clearly aiming for convergent validity.
- [ ] By making sure that the outcome of the qualitative observation can be reorganized into a quantified measurement.
- [x] By triangulation - using multiple methods of observation about the same property.

12: Why would one want to perform a case study? Mark all correct options.

- [x] Because multivariate studies are often unable to tell us about cause-effect relationships in an individual case, while case studies are able to generate such results.
- [x] Because case studies can give us crucial information concerning the boundaries of a particular population.
- [x] Because case studies can give us data that can act as foundations for future studies of highly complex phenomena.
- [ ] Because experimental artefacts cannot occur in case studies.
- [x] Because the results of a case study can be used to falsify a statement which previously was considered to be true.
- [ ] Because case studies can by themselves easily generalise their outcomes to wider populations.

13: What is an analytical narrative?

Using the analytical-narrative method is, in one sense, similar to using the H-D method. You select cases and then create a narrative matching your research question, in order to derive one or several implications (observable consequences) in these cases, under the assumption that your narrative is true. You then test whether these implications are correct to either strengthen or decrease confidence in your narrative. True or false?

- [x] True
- [ ] False

14: A case study is never a better method than multivariate studies; it is just that we often do not have the data to perform the latter. True or false?

- [ ] True
- [x] False

15: There are some common misunderstandings about case studies. Which of the following claims are true? Mark all correct options.

- [x] Case studies, as opposed to multivariate studies, are able to identify cause-effect relationships for individual situations.
- [x] Case studies can help to achieve greater explanatory knowledge.
- [ ] Case studies do not allow us to infer methodologically valid inferences.
- [ ] Case studies cannot utilize quantitative methods.
- [ ] The results from case studies, even if they are valid, cannot be generalized to a context outside of a given case.

Quiz 8: Algorithmic reasoning and its limitations

1: Suppose someone gives you two numbers a and b, and asks you to figure out whether a adjoins b. And suppose you try to answer this by running through all pairs of numbers (x,y) until you find a pair such that ax=by+1. Would this be an adequate method?

- [ ] No. Granted, this method will produce a negative answer if a does not in fact adjoin b; but if a does adjoin b, the investigation will run on forever without ever producing an answer.
- [ ] No; whatever numbers a and b you try this method on, the investigation will invariably run on forever without producing an answer.
- [x] No. Granted, this method will produce an affirmative answer if a does in fact adjoin b; but if a does not adjoin b, the investigation will run on forever without ever producing an answer.
- [ ] Yes; this method is guaranteed to settle the matter.

2: In Section I, a distinction is drawn between a concept's being well-defined and its being effectively decidable. Which of the following best sums up the main message of Section I? Mark the correct option.

- [ ] It is easily seen that all well-defined concepts pertaining to natural numbers are effectively decidable.
- [x] In fact, not all well-defined concepts pertaining to natural numbers are effectively decidable, but showing this will take a bit of work.
- [ ] It is easily seen that not all well-defined concepts pertaining to natural numbers are effectively decidable.
- [ ] In fact, all well-defined concepts pertaining to natural numbers are effectively decidable, but showing this will take a bit of work.

3: Which of the following constructions will yield the (modified) difference function $\dot{-}$ of Example 6.8?

- [ ] Cn[Pr[id, Cn[pred, id_3^3],id, id].
- [x] Pr[id, Cn[pred, id_3^3]].
- [ ] Pr[Cn[id, Pr[pred, id_3^3],id, id].
- [ ] Cn[id, Pr[pred, id_3^3]].

4: Which of the following constructions will yield the (modified) predecessor function of Example 6.7? Mark the correct option.

- [ ] Pr[Cn[z, id_2^3], id, id].
- [ ] Cn[z, id_1^2].
- [ ] Pr[z, id_1^2].
- [x] Cn[Pr[z, id_2^3], id, id].

5: Define f (x, y) = x $\dot{-}$ y, where $\dot{-}$ is as specified in Example 6.8. Then Mn[f](2) is ...

- [x] 2.
- [ ] 1.
- [ ] Undefined.
- [ ] 0.

6: Which of the following statements is true? Mark the correct option.

- [ ] Not every primitive recursive function is total, and not every recursive function is total.
- [ ] Not every primitive recursive function is total, but every recursive function is total.
- [x] Every primitive recursive function is total, but not every recursive function is total.
- [ ] Every primitive recursive function is total, and every recursive function is total.

7: Let f be the characteristic function of the set of perfect numbers. What is f(28)?

- [x] 1.
- [ ] 28.
- [ ] 0.
- [ ] 5.

8: Which of the following is ruled out by Church's thesis? Mark the correct option.

- [x] A total function that is effectively computable but not recursive.
- [ ] A total function that is recursive but not primitive recursive.
- [ ] A total function that is well-defined but not effectively computable.
- [ ] None of the above; they are all compatible with Church's thesis.

9: Suppose someone were to raise the following objection to the differentiation example on the lecture slide entitled Some examples of recursive functions on strings:

"That's not a well-defined function! Look at the first equation. Since $16 = 2^4$, it would mean that $F(16) = F(2^4) = 4 * 2^3 = 32$; but since $16 = (-2)^4$, it would also mean that $F(16) = F((-2)^4) = 4 * (-2)^3 = -32$. Contradiction!"

How would you respond to this objection? Mark the correct option.

- [x] The objector is trying to apply the function F to numbers. This is a mistake, since F is defined as a function that applies to strings of symbols.
- [ ] Good point!
- [ ] The fact that 16 = (-2)4 is irrelevant since -2 is not a natural number.

10: Which of the following holds? Mark the correct option.

- [x] For every n-place recursive function f, there exist several natural numbers a such that $f = \varphi_a^n$.
- [ ] For every n-place recursive function f, there exists exactly one natural number a such that $f = \varphi_a^n$.
- [ ] None of the above: for some recursive functions f, there is no natural number a such that $f = \varphi_a^n$.

11: Which of the following holds? Mark the correct option.

- [ ] All well-defined functions on natural numbers are effectively computable, and all well-defined functions on strings of symbols are effectively computable.
- [x] Not all well-defined functions on natural numbers are effectively computable; nor are all well-defined functions on strings of symbols effectively computable.
- [ ] Not all well-defined functions on natural numbers are effectively computable, but all well-defined functions on strings of symbols are effectively computable.
- [ ] All well-defined functions on natural numbers are effectively computable, but not all well-defined functions on strings of symbols are effectively computable.

Quiz 9: Research ethics

1: The statements below are all normative in that they express some kind of prescription, but what kind of prescription is it? Match each statement with the type of prescription it expresses

You must pay your taxes.	Law
You shouldn't rest your elbows on the table.	Etiquette (convention)
You should look at more data before you accept the hypothesis.	Methodology
You must not cause harm.	Morality
You are not allowed to pick up the ball with your hands.	(Non-legal) convention

2: What distinguishes the ethical frameworks of consequentialism and deontology from each other? Mark all correct options.

- [x] Unlike deontology, consequentialism focuses solely on consequences when evaluating the moral status of actions.
- [x] Unlike consequentialism, deontology attributes moral status to actions, rather than the consequences of actions.
- [ ] While both consequentialism and deontology are 'consequence-oriented frameworks', consequentialism (unlike deontology) only attributes moral status to the immediate consequences of actions. Deontology also takes their indirect consequences into account.

3: Consequentialism and deontology both make important contributions to research ethics, but in what way? Mark all correct options.

- [x] Deontology respects rights and explains why credit attribution needs to be done carefully.
- [x] Deontology emphasises that harming another human is bad, no matter what.
- [x] Consequentialism factors in the epistemic goals of science (good predictions, for instance).
- [x] Consequentialism emphasises the greater good, thus factoring in the effects of a scientific study not only on the particular subjects, but on the whole scientific community, as well as on humans in general.

4: What is virtue ethics about? Characterize virtue ethics by the claims below.

- [x] Morality is not (primarily) about duties and consequences.
- [ ] A virtue-ethical framework consists of a list of duties that all humans are morally obliged to follow.
- [x] You learn how to be a virtuous person by attempting to behave like other virtuous persons.
- [x] Morality consists in the possession and exemplification of good character traits.

5: You have now learned about three frameworks of normative ethics. What are their characteristics? Match each statement to the right moral framework.

If Ann has told a lie that had owerwhelmingly bad consequences and Bob defends Ann by showing that her intentions were the best, then Bob's defense is appealing to...	virtue ethics
To act morally is to fulfill relevant rules and duties.	deontology
If I aim to act in a way that will maximize the happiness in the world, then I am trying to follow a certain kind of...	consequentialism
Morality consists in having and developing good character traits.	virtue ethics
Actions and choices are to be morally assessed only by their consequences.	consequentialism

6: Which part of the principle of informed consent is threatened by which choice of experimental setup? Match each description to the criterion that it most directly threatens.

The research subjects are offered ten times their daily salary for each day's participation.	Voluntary
During a large part of the study, the research subjects will be isolated in a closed room with no means of communicating with the researchers.	Decisionally capacitated
The research subjects are blinded by not knowing the purpose or the setup of the study.	Informed

7: Which of these are examples of consequentialist arguments for using the principle of informed consent? Mark all correct options.

- [x] If people are allowed to decide for themselves, people's trust in science increases, which makes them more likely to participate in scientific studies, which could help save lives.
- [x] If people are allowed to decide for themselves, better consequences are generated, since we all know what makes ourselves happy better than anyone else does.
- [ ] If people are allowed to decide for themselves, people's rights to their own bodies are respected and that is one fundamental right that cannot be violated.

8: In the video it is argued that the Volkswagen case, as it is presented, is problematic from a virtue ethical perspective. Suppose that the case were different in one of the ways below. Which alteration of the scenario would make it unethical from a consequentialist perspective? Which would make it unethical from a deontologist perspective? Match the descriptions with the right framework (the one for which it is most clearly a problem).

The participants are not well-educated and are in great need of the money they are paid for participating.	Unethical according to deontology
The participants are not informed that a previous study showed some negative health effects of exposure to the pollutants.	Unethical according to deontology
Due to unforeseen events beyond the researchers' control, the experiment causes great harm to the participants, while adding very little to scientific progress.	Unethical according to consequentialism

9: Which of the following statements about the morality of fabricating and falsifying data are correct?

- [x] Fabricating or falsifying data could produce bad consequences, such as reducing the public's trust in science.
- [x] Fabricating or falsifying data is dishonest and exhibits poor character traits.
- [x] To fabricate or falsify is to lie, and lying is (typically) deontologically prohibited.
- [ ] Falsification isn't morally blameworthy since we can never draw the line between falsification and acceptable manipulation of data.
- [ ] All bad methodology is morally blameworthy, and since both fabrication and falsification are examples of bad methodology, they are always morally blameworthy.

10: How can we mitigate the effects of scientific misconduct in the form of fabrication, falsification or plagiarism? Mark all correct options.

- [x] We should support the peer-review system and expose wrongdoers.
- [x] We should counteract such conduct by requiring transparency in scientific institutions.
- [x] We should impose continued scrutiny of findings, perhaps by increasing repetition and replication of experiments.
- [ ] We should trust that the problem resolves on its own.

11: According to the 2003 definition by the ICMJE (International Committee of Medical Journal Editors), what are the criteria that someone needs to fulfil in order to be the author of a publication? Mark all correct options.

- [ ] Performing the work alone.
- [x] Agreeing to be fully accountable for all aspects of the work.
- [x] Substantial contribution to the conception (the creation) of the work.
- [x] Final approval of the work.
- [ ] Contributing to the funding of the work reported on in the publication.
- [x] Drafting the final work.

12: What are the ethical reasons to say that correct attribution of authorship is important? This problem builds on the previous lectures about the normative ethical theories. Now it is time to apply them to the question of authorship. Match each argument below with the normative theory it is connected with in the lecture.

It respects intellectual property rights.	Deontology
It is a way of saying that one must always take responsibility for one's work.	Deontology
It provides incentives for scientists, in extension leading to more fruitful science.	Consequentialism
It is the way in which a good scientist behaves.	Virtue ethics
It increases the public's trust in science.	Consequentialism

13: Which of the following moral concerns should researchers weigh in when conducting a study?

- [x] Their own interests and inclinations.
- [x] The efficiency of their efforts.
- [x] The promotion of the total good by the study.
- [x] Their intentions with the study.

14: Match the different normative theories with the description of how they could weigh in on the choice of a career.

One should choose a career so that one fulfills the rights and duties of humans, without making oneself a means to improving the world.	Deontology
One should choose a career so that one flourishes in one's virtues, even if this means spending your life on other things than for instance ending world hunger.	Virtue ethics
One should choose a career in which one creates the best consequences for the largest amount of people, and if necessary disregard any personal interests or preferences.	Consequentialism

15: Now that you have watched all videos on ethics, which of the following statements about ethics in science are true? Mark all correct options.

- [x] The three ethical frameworks sometimes result in different verdicts. One example of this is the question of moral career choices.
- [ ] Out of the three ethical frameworks, deontology is the only one that can be used to argue for the principle of informed consent in science.
- [x] In order to do a good ethical analysis of a scientific project we often need to apply more than one ethical framework.
- [ ] Scientific misconduct is not unethical, it is just methodologically bad.
- [x] Ghost and gift authorship are arguably unethical practices according to all three ethical frameworks.

Quiz 10: Anticipating risk in science and engineering

1: Match each qualitative risk with the correct definition of risk.

Playing the world cup final is risky - it may either make you a satisfied winner or bitter loser.	The spread of events
Poisoning from contaminated water is a substantial risk of unregulated well drilling.	An unwanted event
Building cheap and uninspiring apartment blocks that may make people feel depressed poses a risk.	The cause of an unwanted event

2: Match each quantitative risk with the correct definition of risk.

Either no one dies in the treatment group or everyone does, because they are all subjected to the same intervention.	The severity of an unwanted event
A professional boxing career carries many risks: you might break one or more of your ribs, but you might also get traumatic brain injuries.	The severity of an unwanted event
The risk of getting cancer from smoking is twice as big as the risk of getting cancer from spending too much time at the solarium.	The probability of an unwanted event

3: Which of the following examples contain a correct characterization of subjective probabilities? Mark all correct options.

- ☐ Subjective probability is the frequency with which event E occurs.
- ☑ The subjective probability of an event E can be assessed by considering what kind of bets an individual agent would be willing to participate in.
- ☑ When someone says, since the opposing team's key player is missing we will probably win today, it marks a subjective probability assessment.
- ☐ An example of a subjective probability assessment is when I calculate the probability that a fair (six-sided) dice will yield the value "three" when I throw it one time.

4: Which of the following claims are correct regarding decision-making under certainty/uncertainty? Mark all correct options.

- ☑ Deep uncertainty is when one lacks knowledge about what all possible outcomes are.
- ☐ If we know the probabilities of some possible outcomes, but lack knowledge about what all possible outcomes are, we are in a situation of decision-making under ignorance.
- ☐ In a decision under certainty there is a small – practically non-existent – risk that the believed outcome does not occur.
- ☑ A necessary requirement for something to be a decision-making under risk is that the probability of each possible outcome is known.

5: Which of the following claims are true about the notion of the statistical expected value of an uncertain event? Mark all correct claims.

- ☑ The notion of the statistical expected value combines different quantitative notions of risks in order to give each random variable a numerical value that is supposed to represent the probability-weighted average of all its possible values.
- ☑ Depending on the context, the aim is to choose the action with the lowest expected value or the action with the highest expected value, when applying this notion of risk.
- ☐ According to this notion, the actual risk is the event that will occur (even though we are uncertain what it might be).
- ☐ To quantify the statistical expected values of actions involving heterogeneous outcomes never leads to issues of comparability.

6: What is the difference between risk assessment and risk management?

- ☑ Risk assessment is ideally free from values (although, practically, this is debatable), whereas risk management is laden with values.
- ☑ Risk assessment concerns the identification and evidential justification of models of the risks, whereas risk management concerns the evaluation of possible implications from the available actions and their risks.
- ☑ Risk management is arguably the policy maker's area of expertise, whereas risk assessment is arguably the scientist's area of expertise.
- ☑ Political institutions typically should deal with risk management, and not with risk assessment.

7: Consider a case where the statistical expected values between two or more actions are the same. What problem(s) does such a case present? Mark all correct options.

- [ ] Such cases show that it's not necessary to define 'risk' as 'statistical expected value'.
- [ ] Such a case does not really present a problem because the statistical expected values of two risks can never be the same value.
- [x] Such cases show that it's not sufficient to define 'risk' as 'statistical expected value'.
- [x] In such a case, the notion of the statistical expected value can't by itself guide one's decision.

8: What does the St. Petersburg paradox show? That is, what conclusions could be drawn from the St. Petersburg paradox? Mark all correct options.

- [x] The paradox shows that the expected-value model of risk does not account for the possibility that the utility of something valuable (e.g. money) is diminishing in proportion to its magnitude.
- [x] The paradox shows that, sometimes, the expected-value model is unusable without incorporating subjective evaluations of risk.
- [x] The paradox shows that, depending on what is being quantified, people's tendency to be risk averse/risk-loving relative to what is valued ought to be accounted for.
- [ ] The paradox shows that the expected value of risky actions is sometimes different from what the expected value model says.

9: Should psychological properties that clearly influence subjective evaluations be included in risk evaluation? Mark the correct option.

- [ ] No, such properties are irrelevant for objective risk evaluations.
- [x] Experts currently disagree on how to answer this question.
- [ ] Yes, such properties need to be taken into account in risk evaluations.

10: What are some unwanted consequences if one were to apply the precautionary principle too strictly and without taking context-specific considerations into account?

- [x] It could risk creating too strict thresholds that in the end create paralysis among decision makers and proposing parties.
- [x] It would yield the opportunity for some decisions maker(s) to negate projects based on insufficient evidence, while the real reason is that the projects aren't in line with the respective decision maker(s)' personal interests.
- [x] It could risk neglecting important projects that indeed are backed up by evidence pointing to the absence of a risk or its relatively low probability, by demanding "proofs".

11: Why might one want to model risks using qualitative models rather than quantitative models?

- [x] From an epistemic viewpoint, the accuracy of quantitative models is sometimes too questionable to guide important decisions.
- [x] From a public viewpoint, we sometimes want as many people as possible to grasp a certain model of a risk.
- [ ] From an ethical viewpoint, we might want to assess the severity and probability of the possible outcomes of some decision.
- [x] From a pragmatic viewpoint, we sometimes lack resources or time to model some risks quantitatively.

12: Match each description with its corresponding decision principle.

Due to a lack of research, company C decides that they will weigh all the possible outcomes of action A equally when calculating A's expected value.	Principle of insufficient reason
When company C proposes to build a new power plant next to a lake, the government asserts that they will only allow this construction if C can provide conclusive evidence that the plant will not contaminate the lake.	Precautionary principle
During the heated moments of the cold war, nuclear devastation was so close that each action taken by the Americans and the Russians was based on the goal that it would not lead to a nuclear attack.	Minimax

13: Match the possible approaches that a utilitarian could utilize when dealing with risk management.

Given that enough information is available and because one, arguably, would want to have some idea of an action's utility before one possibly performs the action, an idea could be to extract an action's utility via the statistical-expected-value model.	The expected utility approach
Since utilitarianism exclusively values the utility (consequences) of actions, one could interpret an uncertain action's utility to be the utility of the outcome that happens, once the action is performed. This way, we get a clear criterion for deciding whether an action was right or wrong, but it can only be done once the action has been taken.	Actualism
A form of utilitarianism is rule utilitarianism, i.e. the right action is the action that maximizes utility in accordance with some pre-determined normative rules. Utilitarianism could apply such rules to guide risk management.	None of the explicit approaches

14: What is the point of Nozick's critique, i.e. what does his critique show? Mark all correct options.

- ☐ Nozick's critique shows that, in risk management, a normative framework must take into account the managing person's virtues. In other words, is the managing person acting with a "good heart"?
- ☑ Nozick's critique targets normative frameworks whose underlying principles rest on the view that violating someone's right(s) is unconditionally morally wrong. He shows that such frameworks get into trouble when actions have to be chosen under risk, ignorance, and deep uncertainty.
- ☐ Nozick's critique shows that the consequences of exposing someone to uncertain risks are worse than exposing them to certain outcomes (even if such outcomes are negative).

15: Match the possible approaches that a deontologist could utilize when dealing with risk management.

Arguably, one's fate is something one choses for oneself. Therefore, if one chooses to participate/expose oneself in a situation where some risk of harm is present, the deontologist can argue that someone creating this risk is not acting wrong.	The "voluntary approach" is an available approach for deontology
The deontologist could make sure that when managing risks, she manages these risks with great care and attention to the reason(s) that drives her management – are there any of her character traits that seem to guide her to unwarranted decisions?	Not an available approach for deontology
Intuitively, there is a difference when, say, action A1 is taken with the intention to create outcome O1 while A2 is taken without the intention to create O2 (even though O1 and O2 are both negative outcomes). A deontologist could utilize this distinction of a subject's intention(s) to distinguish between morally permissible actions and morally forbidden actions.	The "intention approach" is an available approach for deontology